# WORKSHEET 3.4 · MINIMUM DISTANCE

11a) $$d = \frac{|3(3) - 2(-2) - 6|}{\sqrt{3^2 + (-2)^2}}$$

$$= \frac{|9 + 4 - 6|}{\sqrt{13}}$$

$$\boxed{d = \frac{7}{\sqrt{13}} \text{ or } \frac{7\sqrt{13}}{13}}$$

b) $$\frac{x-3}{2} = \frac{y-4}{7}$$

$$7(x-3) = 2(y-4)$$

$$7x - 21 = 2y - 8$$

$$7x - 2y - 13 = 0$$

$$d = \frac{|7(3) - 2(-2) - 13|}{\sqrt{7^2 + (-2)^2}}$$

$$= \frac{|21 + 4 - 13|}{\sqrt{53}}$$

$$\boxed{d = \frac{12}{\sqrt{53}} \text{ or } \frac{12\sqrt{53}}{53}}$$

c) $$\vec{r} = (-3, -7) + t\left(\frac{1}{5}, \frac{1}{6}\right)$$

$$\vec{m} = \left(\frac{1}{5}, \frac{1}{6}\right) \text{ or } 30\left(\frac{1}{5}, \frac{1}{6}\right) = (6, 5)$$

$$\vec{n} = (5, -6)$$

$$5(-3) - 6(-7) + C = 0$$

$$-15 + 42 + C = 0$$

$$C = -27$$

$$\therefore 5x - 6y - 27 = 0$$

$$d = \frac{|5(3) - 6(-2) - 27|}{\sqrt{5^2 + (-6)^2}}$$

$$= \frac{|15 + 12 - 27|}{\sqrt{61}}$$

$$d = \frac{0}{\sqrt{61}} = 0$$

$\therefore$ Point $Q(3, -2)$ is on the line !!!

d) $x = -5$

$x + 5 = 0$

$$d = \frac{|(3) + 5|}{\sqrt{1^2 + 0^2}}$$

$$\boxed{d = 8}$$

12 a) $$d = \frac{|6(4) + 3(7) - 10|}{\sqrt{6^2 + 3^2}}$$

$$= \frac{|24 + 21 - 10|}{\sqrt{45}}$$

$$= \frac{35}{3\sqrt{5}}$$

$$\boxed{d = \frac{7\sqrt{5}}{3}}$$

b) $$d = \frac{|6(4) + 3(-8) - 10|}{\sqrt{6^2 + 3^2}}$$

$$= \frac{|24 - 24 - 10|}{\sqrt{45}}$$

$$= \frac{|-10|}{3\sqrt{5}}$$

$$\boxed{d = \frac{2\sqrt{5}}{3}}$$

c) $$d = \frac{|6(0) + 3(5) - 10|}{3\sqrt{5}}$$

$$= \frac{5}{3\sqrt{5}}$$

$$\boxed{d = \frac{\sqrt{5}}{3}}$$

d) $$d = \frac{|6(5) + 3\left(\frac{-20}{3}\right) - 10|}{3\sqrt{5}}$$

$$= \frac{|30 - 20 - 10|}{3\sqrt{5}}$$

$$\boxed{d = 0.}$$

15a) see lesson notes

b) $$d = \frac{|\vec{PQ} \times \vec{m}|}{|\vec{m}|}$$

$\vec{m} = \langle 1, 1, 2 \rangle$

$\vec{PQ} = (-2, -3, -3)$

$$\begin{vmatrix} -2 & -3 & -3 & -2 & -3 \\ 1 & 1 & 2 & 1 & 1 \end{vmatrix}$$

$= (-6 + 3, -3 + 4, -2 + 3)$

$= (-3, 1, 1)$

$$= \frac{|(-3, 1, 1)|}{\sqrt{1^2 + 1^2 + 2^2}}$$

$$= \frac{\sqrt{(-3)^2 + (1)^2 + (1)^2}}{\sqrt{6}}$$

$$\boxed{d = \frac{\sqrt{11}}{\sqrt{6}} \text{ or } \frac{\sqrt{66}}{6}.}$$

c) $\vec{r_1} = (-2, 2, 1) + t(7, 3, -4)$ , $t \in \mathbb{R}$

$\vec{r_2} = (2, -1, -2) + u(7, 3, -4)$ , $u \in \mathbb{R}$

Let $P(-2, 2, 1)$

$Q(2, -1, -2)$

$\therefore \vec{PQ} = (4, -3, -3)$

$\vec{m} = (7, 3, -4)$

$$d = \frac{|\vec{PQ} \times \vec{m}|}{|\vec{m}|}$$

$$\vec{PQ} \times \vec{m} = \begin{vmatrix} i & -3 & -3 & 4 & -3 & -3 \\ 3 & \swarrow & -4 & \swarrow & 7 & \swarrow & 3 & -4 \end{vmatrix}$$

$$= (12 + 9, -21 + 16, 12 + 21)$$

$$= (21, -5, 33)$$

$$= \frac{|(21, -5, 33)|}{\sqrt{7^2 + 3^2 + (-4)^2}}$$

$$= \frac{\sqrt{(21)^2 + (-5)^2 + (33)^2}}{\sqrt{74}}$$

$$= \frac{\sqrt{1555}}{\sqrt{74}}$$

$$= \frac{\sqrt{115070}}{74}$$